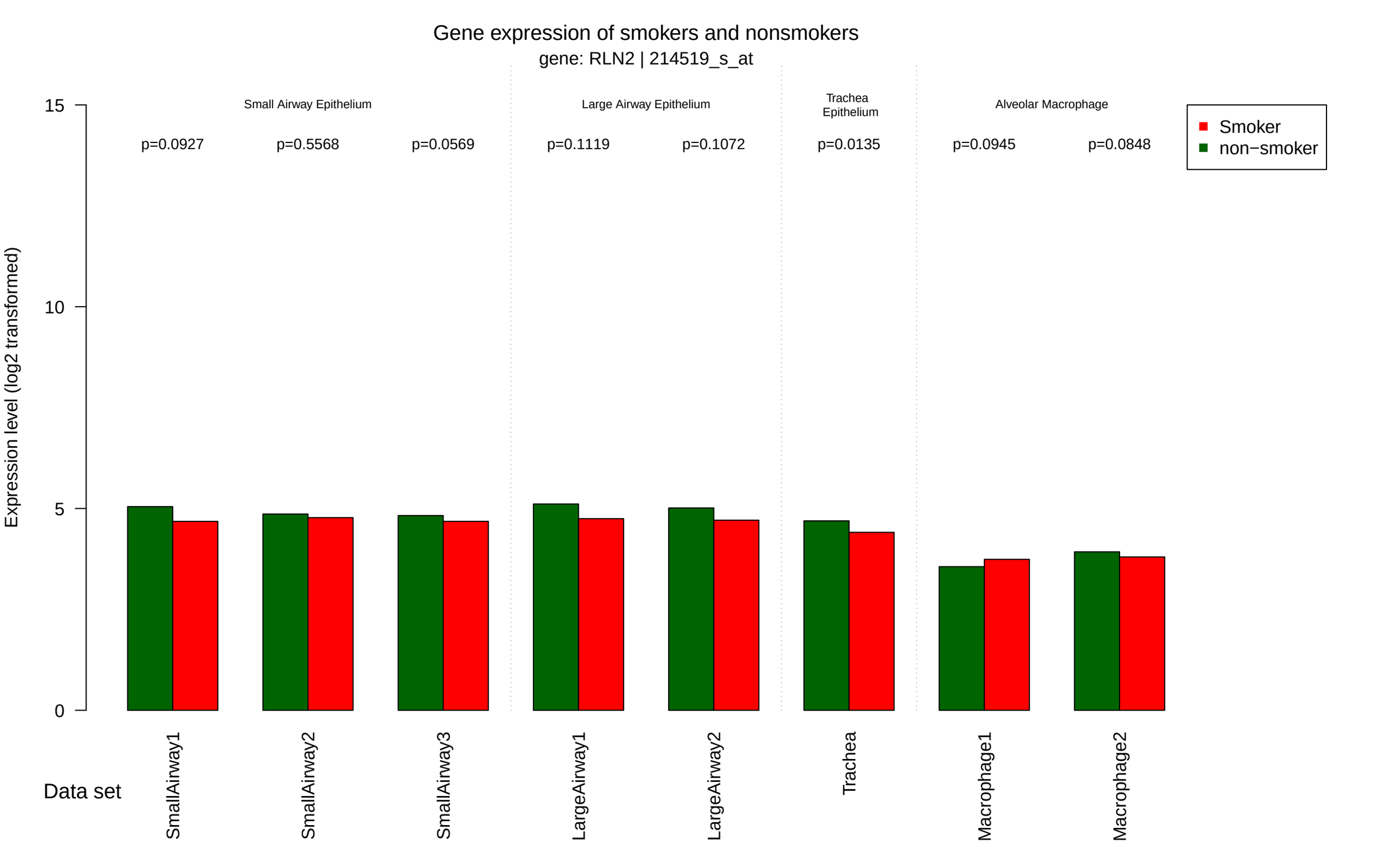
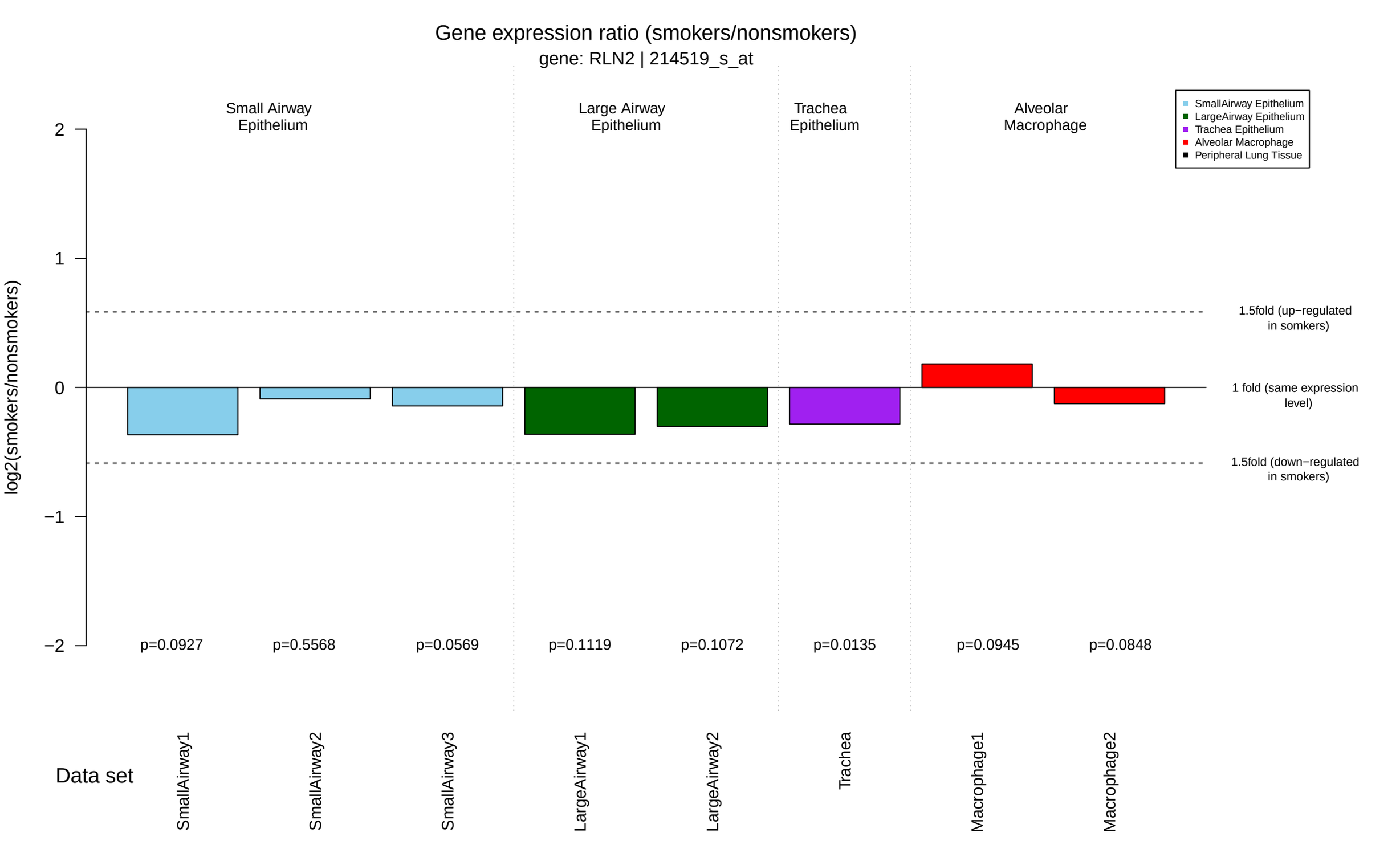
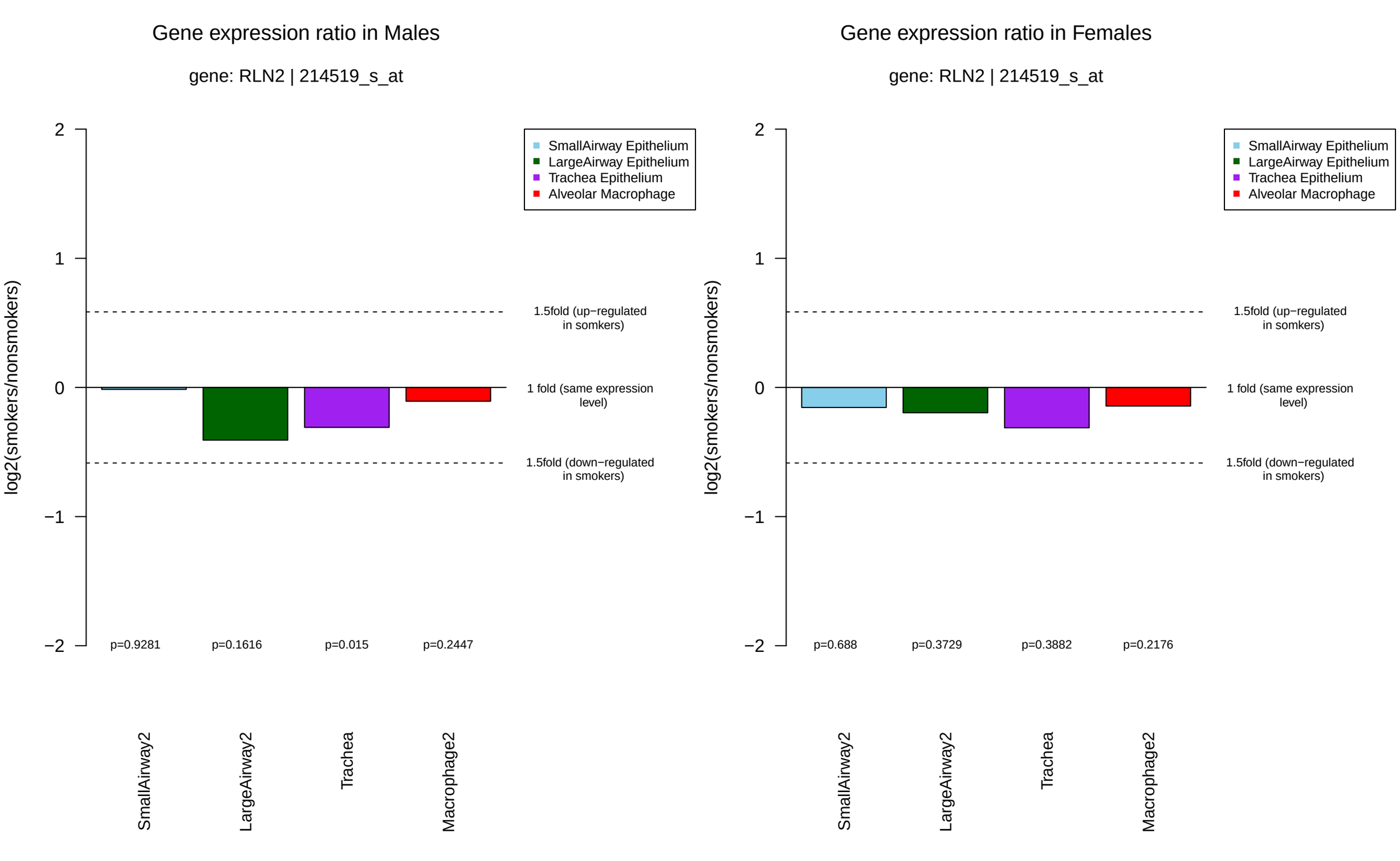
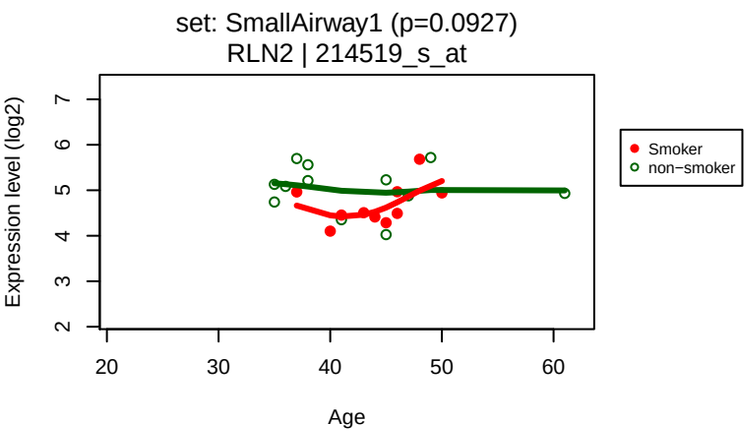
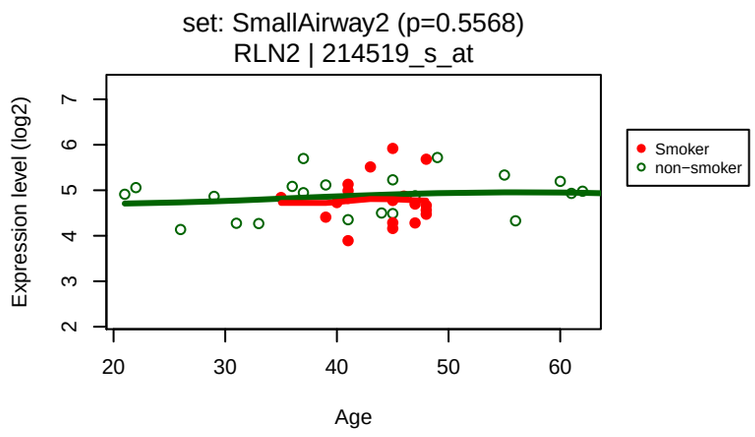
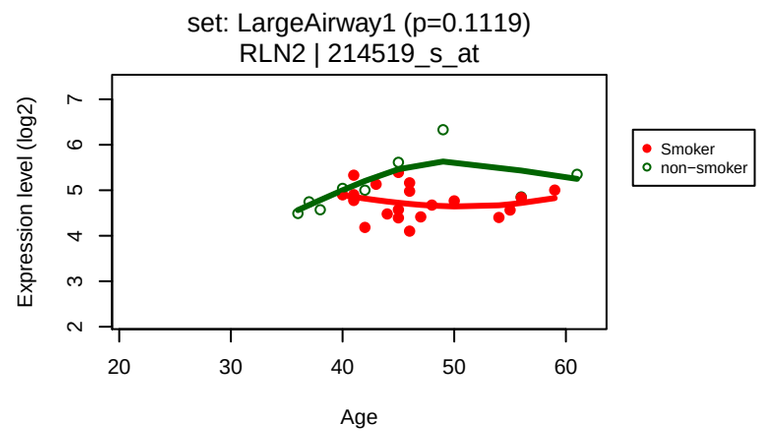

Gene expression of smokers and nonsmokers
gene: RLN2 | 214519_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0927
p=0.5568
p=0.0569
p=0.1119
p=0.1072
p=0.0135
p=0.0945
p=0.0848
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

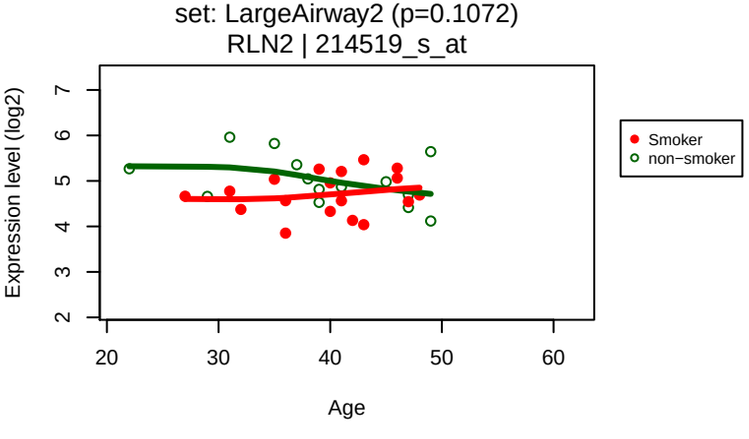

Gene expression ratio (smokers/nonsmokers)
gene: RLN2 | 214519_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0927
p=0.5568
p=0.0569
p=0.1119
p=0.1072
p=0.0135
p=0.0945
p=0.0848
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

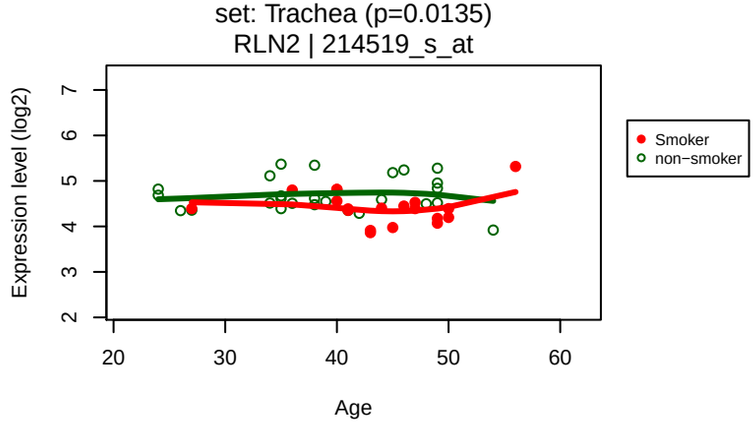

Gene expression ratio in Males
gene: RLN2 | 214519_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9281
p=0.1616
p=0.015
p=0.2447
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RLN2 | 214519_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.688
p=0.3729
p=0.3882
p=0.2176
SmallAirway2
LargeAirway2
Trachea
Macrophage2

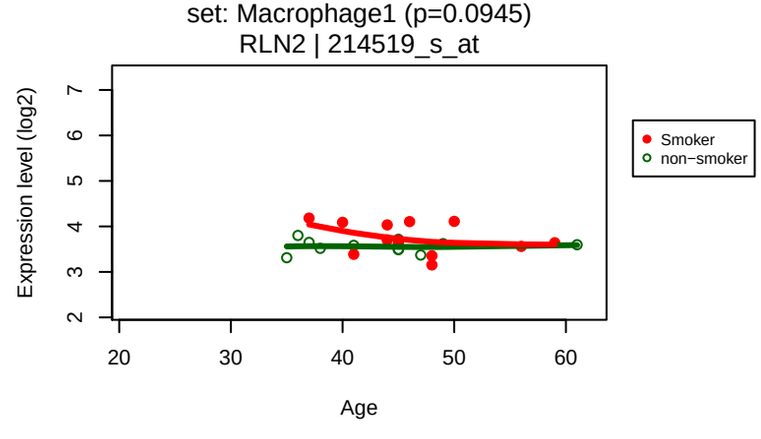

set: SmallAirway1 (p=0.0927)
RLN2 | 214519_s_at
Expression level (log2)
Age
Smoker
non−smoker

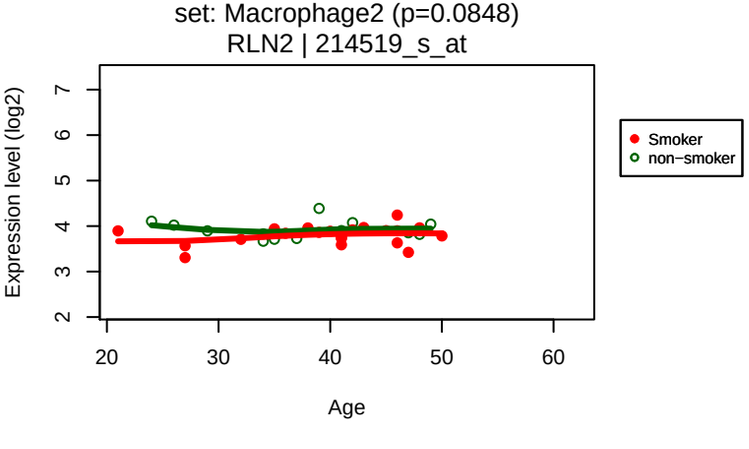

set: SmallAirway2 (p=0.5568)
RLN2 | 214519_s_at
Expression level (log2)
Age
Smoker
non−smoker

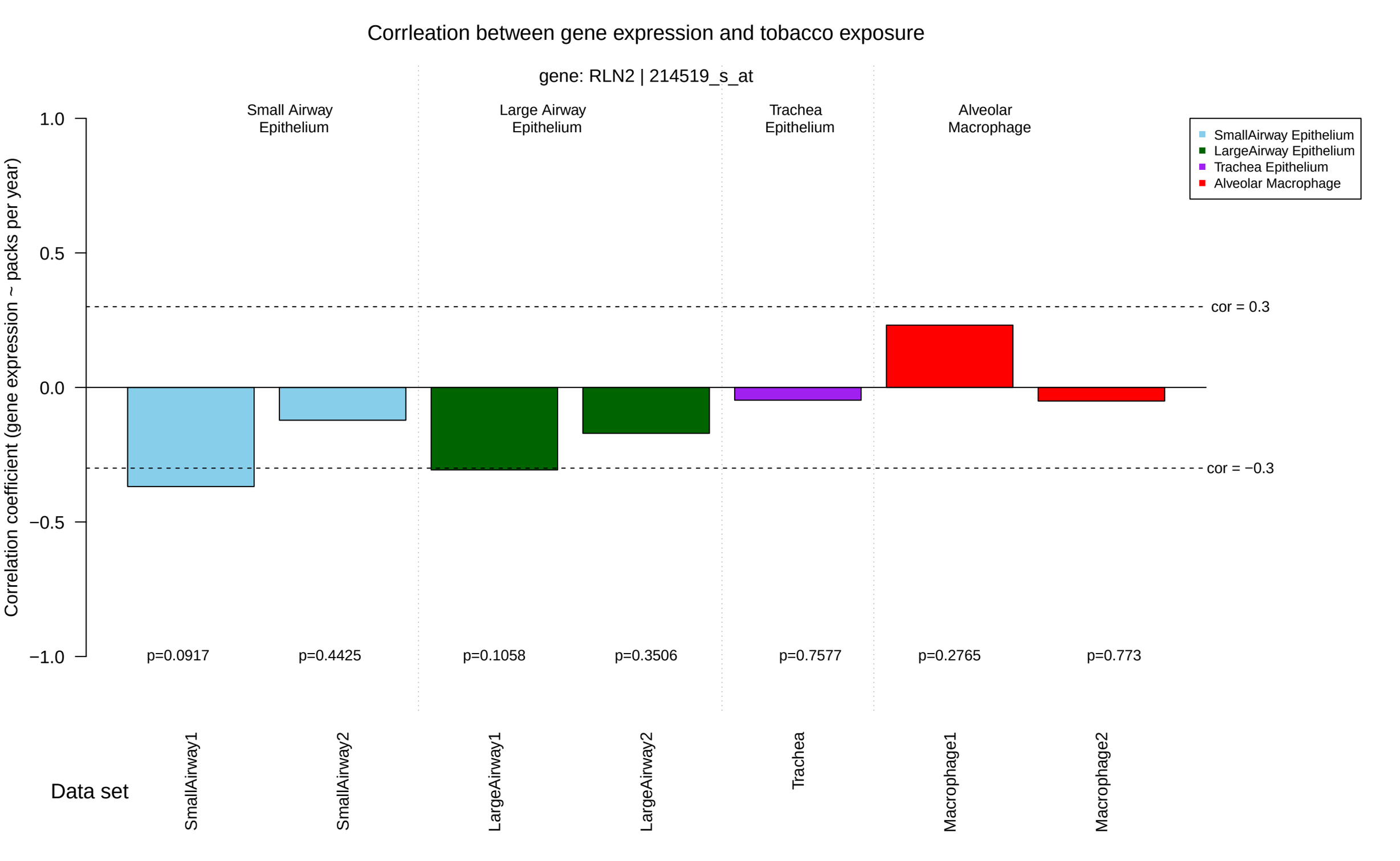

set: LargeAirway1 (p=0.1119)
RLN2 | 214519_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1072)
RLN2 | 214519_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0135)
RLN2 | 214519_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0945)
RLN2 | 214519_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0848)
RLN2 | 214519_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RLN2 | 214519_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0917
p=0.4425
p=0.1058
p=0.3506
p=0.7577
p=0.2765
p=0.773
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2